
Model-Error-Driven Contrastive Fine-Tuning for Robust Hate Speech Detection

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Abstract

Automated hate-speech classifiers often exploit surface-level lexical cues rather than genuine hateful intent. In particular, identity-related terms (*Muslim*, *Black*, *gay*) become spurious shortcuts during training, causing non-hateful identity-related statements to be misclassified. We propose a model-error-driven contrastive fine-tuning pipeline: false positives from a RoBERTa baseline are analysed with Integrated Gradients to identify identity trigger terms, then paired with minimally rewritten hateful counterparts to fine-tune with a mixed contrastive-cross-entropy objective. Across 208 hyperparameter configurations on HateXplain, a balanced setting reduces test identity false positives from 26 to 22 with a 1.9-point recall drop; an aggressive setting brings them to 17 at the cost of a 5.2-point recall drop. Attribution analysis confirms reduced reliance on identity terms, though attribution does not consistently shift toward hateful-context words. The method provides a controllable false-positive/false-negative trade-off and a principled framework for targeted lexical bias mitigation.

Keywords: hate speech detection, contrastive learning, identity bias, false positives, RoBERTa.

1. Introduction

Automated hate speech moderation is deployed at massive scale, yet it carries a specific and consequential failure mode: over-blocking legitimate discussion of minority groups introduces systematic bias against the very communities these systems are meant to protect. A classifier that treats *Muslim* or *gay* as a signal for hate will silence counterspeech, news reporting, and community conversation alongside genuine hate. This false-positive failure is not only harmful to affected communities but also erodes trust in automated moderation systems.

We address the following research question: *Can model-*

error-driven contrastive fine-tuning reduce false positives on identity-related non-hateful text without substantially degrading recall on genuinely hateful content? The key design principle is to start from the classifier’s own validation errors rather than applying generic data augmentation—making the intervention both data-efficient and directly grounded in the observed failure mode. We demonstrate that the pipeline can be controlled through three hyperparameters, allowing practitioners to choose the operating point that fits their moderation trade-off.

2. Motivation and Related Work

Our work builds directly on HateXplain [1], which introduced token-level human rationales alongside hate speech annotations. A key finding is that transformer models can associate identity terms with hateful labels even in benign contexts. While HateXplain’s rationales help diagnose where the model attends, rationale-based training alone does not directly reduce identity-related false positives. We take this gap as our starting point and show in Sec. 5 that a strong RoBERTa baseline still produces many such errors.

To address this gap, we seek a training signal that teaches the model to distinguish neutral from hateful uses of the same identity term. Counterfactual data augmentation (CDA) is a natural candidate, but Sen et al. [2] show that when pairs are generated without reference to the model’s actual failure distribution, CDA can increase identity-related false positives by targeting the wrong cases.

This motivates our use of contrastive learning [3]. A contrastive objective can push apart representations of examples that share identity cues but differ in label, teaching the model that the same identity term can appear in both hateful and non-hateful text. Unlike standard contrastive approaches, we source pairs from the model’s own false positives and use Integrated Gradients to confirm that identity terms drive the error before writing each pair. The intervention is therefore model-error-driven and targeted rather than applied wholesale.

3. Dataset

We use the HateXplain dataset [1] above all because it explicitly separates *hate speech*, *offensive language*, and *normal* content, making it possible to isolate the specific failure mode we target and because each post was annotated by at least three human annotators on Amazon Mechanical Turk. This manual annotation quality was critical for our approach: the entire pipeline rests on trusting that collected false positives are genuinely non-hateful. Noisy or automatically derived labels would corrupt this diagnostic step.

Label construction. We map to binary labels using a strict majority rule over per-annotator votes: *hateful* if the majority voted hate speech, *non-hateful* if the majority voted normal. Posts where the majority voted *offensive* are excluded, since offensive content is not necessarily hateful and keeping it would blur the decision boundary. After filtering, the dataset contains 10,999 / 1,374 / 1,376 examples in the train, validation, and test splits, with a 57/43 non-hateful to hateful ratio. The validation set was used for false positive mining; the test set was held out entirely until final evaluation.

Contrastive pair dataset. We additionally construct 128 targeted contrastive pairs from 64 validation false-positive candidates; Sec. 4 details the construction process.

4. Method

Step 1 — Baseline model. We fine-tune RoBERTa-base [4] for binary hate speech classification, with a linear head over the [CLS] token (RoBERTa’s beginning-of-sequence token, whose final hidden state aggregates the full sequence for classification). Training uses cross-entropy for 3 epochs (batch size 16, learning rate 2×10^{-5} , weight decay 0.01), selecting the checkpoint with the best validation Macro-F1. The baseline reaches test Macro-F1 0.895, with 70 false positives and 26 identity false positives.

Step 2 — Identity term discovery and false positive selection. We run inference on the validation set and collect all 71 false positives (gold: non-hateful, predicted: hateful). We then apply Integrated Gradients to the hateful logit of each false positive, aggregating subword attribution scores to word level and filtering out stop words to retain only semantically meaningful trigger terms. Starting from a seed list of 15 common identity terms (*Muslim*, *Black*, *Jewish*, *gay*, etc.), this attribution pass surfaces slurs and variant spellings the model had learned to over-trigger on but that were absent from the seed list. After deduplication and manual review, the final vocabulary contains 53 terms and matches 64 of the 71 false positives, which become the contrastive pair candidates.

Step 3: Contrastive pair construction. Each of the 64

validated candidates is rewritten twice, yielding 128 (source, hateful counterpart) pairs. Hateful counterparts were generated with an LLM and then manually reviewed: each preserves the identity term, approximate length, and sentence structure while shifting the semantic intent to clearly hateful content.

Step 4: Contrastive fine-tuning. Training starts from the baseline checkpoint, as a targeted second stage that adapts a competent classifier rather than retraining from scratch. The mixed loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}^{\text{train}} + \beta \mathcal{L}_{\text{CE}}^{\text{pairs}} + \alpha \mathcal{L}_{\text{con}}, \quad (1)$$

where $\mathcal{L}_{\text{CE}}^{\text{train}}$ is cross-entropy on the full HateXplain training set, $\mathcal{L}_{\text{CE}}^{\text{pairs}}$ is cross-entropy on the 128 contrastive pairs, and $\mathcal{L}_{\text{con}}$ pushes the final-layer $\langle s \rangle$ representations of each pair apart:

$$\mathcal{L}_{\text{con}} = \max(0, m - z_{\text{non}} - z_{\text{hate2}})^2. \quad (2)$$

The margin m sets the minimum embedding separation the loss enforces; if a pair’s distance already exceeds m , $\mathcal{L}_{\text{con}}$ evaluates to zero. Scalar weights α (embedding pressure) and β (pair-classification emphasis) are swept independently over $\alpha \in [0.05, 5.0]$, $\beta \in [0.0, 1.0]$, and $m \in \{1, 5, 10, 15, 18, 20, 22\}$ —208 configurations total.

5. Validation

Experimental setup. We evaluate on the held-out test set (1,376 examples: 782 non-hateful, 594 hateful). Primary metrics are identity false positives (ID-FP), hateful recall, and Macro-F1. No test-set information informed pair selection or vocabulary construction.

Hyperparameter effects. Before the grid search, we computed the L2 distance between baseline $\langle s \rangle$ embeddings for each contrastive pair and found the median already exceeded 5, meaning that with $m \in \{1, 5\}$ the contrastive loss evaluated to zero from the first training step, so α had no effect. This motivated pushing margins up to 22. Once the margin is active ($m \geq 10$), α controls embedding separation pressure: higher values reduce ID-FP but degrade recall, as the model becomes globally more conservative. β proves more targeted: training the model to correctly label the pairs achieves ID-FP reductions with a smaller recall penalty. The best trade-off configurations combine a large margin ($m \in \{18, 20, 22\}$), moderate α (≤ 2.0), and low-to-moderate β (≤ 0.5).

Results. Fig. 1 plots all 208 configurations. Many land to the *right* of the baseline, increasing ID-FP while also hurting recall—this occurs when the contrastive signal is too weak to shift the model’s behaviour on identity terms but still destabilises the classifier. Only configurations with sufficiently large margin and balanced α/β push into the

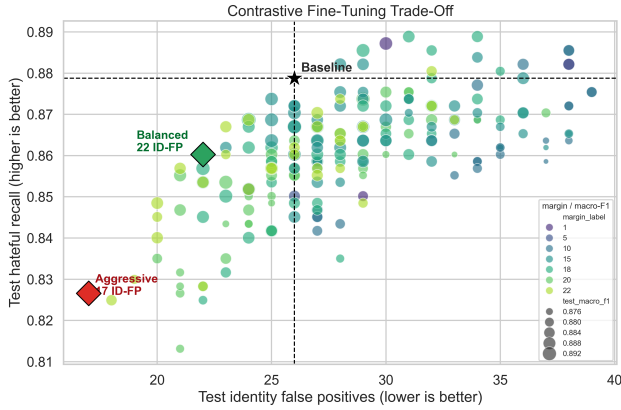


Figure 1. Identity false positives vs. hateful recall across 208 configurations. Color encodes margin; size encodes Macro-F1. Many configurations increase ID-FP relative to the baseline (points to its right), confirming that the contrastive objective is not uniformly beneficial.

Model	ID-FP	FP	FN	ID-FN	Rec.	M-F1
Baseline	26	70	72	59	.879	.895
Balanced	22	63	83	67	.860	.891
Aggressive	17	47	103	87	.827	.887

Table 1. Test results for two representative configurations. Balanced: $\alpha=0.2$, $\beta=0.05$, $m=18$. Aggressive: $\alpha=2.0$, $\beta=0.5$, $m=22$. ID-FN: hateful examples containing an identity/trigger term that were missed.

desirable bottom-left quadrant, tracing a Pareto frontier at $m \geq 18$. Tab. 1 highlights two representative points.

False-negative guardrail. Among the 527 hateful test examples containing an identity or trigger term, identity false negatives rise from 59 (baseline) to 67 (balanced) and 87 (aggressive), with identity hateful recall dropping from 0.888 to 0.873 and 0.835 respectively. Part of the ID-FP reduction therefore reflects overcorrection: the models become more reluctant to predict hate even on genuinely hateful identity-related text.

Attribution shift. On the 14 identity false positives shared across all three models, the mean fraction of absolute attribution assigned to identity/trigger vocabulary decreases from 0.368 (baseline) to 0.349 (balanced) and 0.315 (aggressive). Identity attribution consistently decreases across example subsets; hateful-context attribution does not consistently increase—most mass moves into a broad *other* category.

6. Conclusion

Limitations. The intervention uses only 64 source errors and 128 manually written pairs; two rewrites from one source are not independent. Identity and context vocab-

ularies are manual and incomplete, and Integrated Gradients is diagnostic rather than causal. HateXplain labels reflect majority annotator vote and are inherently subjective: some examples we treat as genuine false positives may correspond to borderline cases where annotator disagreement is high and the model’s hateful prediction is not entirely wrong. We preserved all gold labels without re-verification to avoid introducing our own subjective re-annotation, but this means our false-positive set may contain a small number of label-noise cases. External-dataset generalisation is unknown.

We proposed a model-error-driven pipeline that uses Integrated Gradients attribution to validate trigger terms before constructing targeted contrastive pairs for second-stage fine-tuning. The approach avoids the indiscriminate augmentation that prior work showed can backfire, grounding every pair in a confirmed model error. The balanced setting reduces identity false positives from 26 to 22 with a Macro-F1 loss of only 0.004; the aggressive setting achieves 17 at the cost of a 0.008 drop and a 5.2-point recall reduction. The method provides a *controllable* bias-mitigation mechanism: hyperparameters allow practitioners to choose their operating point, but precision gains come from increased conservativeness, not from better semantic generalisation.

Key directions for future work include expanding pair diversity across multiple datasets, causal interventional probing to replace Integrated Gradients, and threshold calibration as a lightweight alternative to second-stage fine-tuning.

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